

Codefest 6

$$a) I_{col0} = \frac{V_{row0} - V_{col0}}{R[0][0]} = \frac{1 - 0}{1000} = 1mA$$

b) KCL at floating V_{row1} :

$$\frac{V_{row1} - V_{col0}}{R[1][0]} + \frac{V_{row1} - V_{col1}}{R[1][1]} = 0 \Rightarrow \frac{V_{row1} - 0}{2000} + \frac{V_{row1} - V_{col1}}{1000} = 0$$

$$\Rightarrow \frac{V_{row1}}{2} + V_{row1} - V_{col1} = 1.5V_{row1} - V_{col1}$$

KCL at floating V_{col1} :

$$\frac{V_{col1} - V_{row0}}{R[0][1]} + \frac{V_{col1} - V_{row1}}{R[1][1]} = 0 \Rightarrow \frac{V_{col1} - 1}{2000} + \frac{V_{col1} - V_{row1}}{1000} = 0$$

$$\Rightarrow \frac{V_{col1}}{2} - \frac{1}{2} + V_{col1} - V_{row1} = 0 \Rightarrow 1.5V_{col1} - V_{row1} = 0.5$$

$$V_{col1} = 1.5V_{row1} :$$

$$1.5(1.5V_{row1}) - V_{row1} = 0.5 \Rightarrow 2.25V_{row1} - V_{row1} = 0.5$$

$$\Rightarrow 1.25V_{row1} = 0.5 \Rightarrow V_{row1} = \frac{0.5}{1.25} = 0.4V$$

$$V_{row1} = 0.4V$$

$$V_{col1} = \frac{0.5 + V_{row1}}{1.5} = \frac{0.5 + 0.4}{1.5} = 0.6V$$

$$\text{Actual } I_{col0} = \frac{V_{row0} - V_{col0}}{R[0][0]} + \frac{V_{row1} - V_{col0}}{R[1][0]} = \frac{1 - 0}{1000} + \frac{0.4 - 0}{2000} = 1mA + 0.2mA = 1.2mA$$

0.2mA comes from the sneak path which is $V_{row0} \rightarrow R[0][1] \rightarrow V_{col1} \rightarrow R[1][1] \rightarrow V_{col0} \leftarrow R[1][0] \leftarrow V_{row1}$

c) The sneak path can add additional unintended current to the column output which makes the actual current larger than the intended current which will corrupt the matrix vector multiply because the measured output no longer represents the intended cell current.